Replication Plan

Exactly-Solved Model of Light-Scattering Errors in Quantum Simulations with Metastable Trapped-Ion Qubits

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Auto-generated Replication Blueprint

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Contents

Paper Overview

This paper presents an analytically solvable model for light-scattering-induced errors in quantum computing with metastable trapped-ion qubits (e.g., $^{137}Ba^+$ or $^{133}Ba^+$ using optical qubits). When laser beams used for gates scatter off nearby ions, they cause decoherence. The authors derive the Lindblad master equation for this error channel, obtain exact solutions for the density matrix evolution, and quantify gate infidelity as a function of ion spacing, laser parameters, and atomic structure. The key results are analytic formulas validated by numerical integration.

Detailed Workflow

Phase 1: Atomic Structure and Scattering Model

1.1 Define the atomic level structure.

- Metastable qubit levels (e.g., $^2S_{1/2}$ and $^2D_{5/2}$ in Ba^+).
- Excited states mediating the scattering.
- Transition wavelengths, dipole matrix elements, branching ratios.

1.2 Compute photon scattering rates.

- Rayleigh and Raman scattering cross-sections from laser light.
- Dependence on detuning and polarization.

Phase 2: Lindblad Master Equation

2.1 Construct the Lindblad operators for the scattering error channel.

$$\dot{\rho} = -i[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right)$$

2.2 Derive the exact solution for the density matrix evolution.

- For the specific structure of the scattering Lindbladians, the master equation is analytically solvable.
- Express $\rho(t)$ in terms of the initial state and scattering parameters.

Phase 3: Gate Fidelity Calculations

3.1 Compute process fidelity for single-qubit and two-qubit gates.

$$\mathcal{F} = \mathrm{Tr}[\rho_{\mathrm{ideal}} \cdot \rho_{\mathrm{actual}}]$$

3.2 Parameterize infidelity as a function of:

- Gate duration t_g.
- Ion separation d.
- Laser intensity and detuning.
- Solid angle of scattered photon collection.

3.3 Numerical integration for validation (Simpson's rule / RK4 for the master equation).

Phase 4: Parameter Studies

4.1 Sweep ion spacing and plot infidelity vs. d.

4.2 Sweep laser detuning and plot error rates.

4.3 Compare analytic vs. numerical solutions to validate.

4.4 Reproduce all figures from the paper.

Datasets

Dataset / Source	Type	Location / Notes
Atomic data (Ba^+)	Published constants	NIST Atomic Spectra Database `https://www.nist.gov/pml/atomic-spectra-database`
Dipole matrix elements	Published	From atomic structure calculations or literature
All computed data	Synthetic	Generated by the analytic/numerical model

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
Python	≥ 3.8	Primary language
NumPy	≥ 1.21	Matrix operations, linear algebra
SciPy	≥ 1.7	ODE integration, special functions
QuTiP	≥ 4.7	Quantum dynamics, Lindblad master equation solver `https://qutip.org/`
SymPy	≥ 1.10	Symbolic derivation of analytic solutions
Matplotlib	≥ 3.4	Plotting fidelity curves and level diagrams
ARC (Alkali Rydberg Calculator)	latest	Atomic structure data for alkali/alkaline-earth ions `https://arc-alkali-rydberg-calculator.readthedocs.io/`

Hardware Requirements

- Any modern laptop; all computations are lightweight.